

Qimin Feng

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PROFESSIONAL SUMMARY

Robotics Ph.D. candidate focused on bio-inspired underwater robots, embedded control, soft robotic actuation, and flow-aware perception. Builds robot systems from hardware integration and real-time control to experimental validation and ML-based sensing. Strongest in project-driven robotics R&D where mechanical design, embedded systems, sensing, and hydrodynamics meet.

TECHNICAL SKILLS

Programming and Systems: Python, C/C++, MATLAB, Linux, Git, FastAPI/WebSocket, Flask

Robotics and Embedded Hardware: Raspberry Pi, ESP32, Arduino, CAN bus, I2C, PWM; motor drivers, voice-coil actuators, magnetic encoders, IMU, pressure/force sensors, NI DAQ

Robotics and Controls: real-time control, motor/servo/VCA control, sensor-actuator integration, impedance shaping, safety interlocks, experimental testbeds

Experimental and Simulation Analysis: PIV & post-processing, hydrodynamic experiments design, SolidWorks, POD/reduced-order modeling, MuJoCo, reinforcement learning

Perception and Machine Learning: PyTorch, CNN/LSTM/attention, sparse sensing, sensor fusion, physics-informed learning, uncertainty estimation, vortex/flow-feature detection

EDUCATION

Iowa State University, Ames, IA

Ph.D. Candidate, Mechanical Engineering | Expected Dec 2027

Research focus: bio-inspired robotics, underwater locomotion, flow sensing, fluid-structure interaction | GPA: 3.97/4.00

Zhejiang University, Hangzhou, China

B.S., Agricultural and Biosystem Engineering | Jun 2022

GPA: 3.35/4.00

RESEARCH EXPERIENCE

Graduate Research Assistant, Zhong Lab, Iowa State University | Jan 2023 - Present

MantaBot: Bio-Inspired Underwater Robot Control Stack

- Built the embedded software architecture for a manta ray-inspired underwater robot, integrating gamepad control, CAN motors, PWM servos, IMU, pressure sensing, and telemetry.
- Developed field-debugging tools including a FastAPI/WebSocket dashboard, Linux/systemd deployment workflow, Tailscale networking, thermal protection, and actuator safety limits.
- Designed free-swimming ground-effect experiments using onboard torque feedback, depth sensing, and overhead tracking to connect robot behavior with hydrodynamic theory.

Voice-Coil Actuator Control for Soft Robotics

- Implemented dual voice-coil actuator firmware on ESP32 with PWM drive, serial command parsing, Python trajectory generation, and encoder-ready pin allocation.
- Debugged GPIO/JTAG conflicts, boot-state glitches, motor-driver behavior, and wiring constraints to produce a stable multi-actuator soft-robotics control setup.
- Designed an ESP-NOW commander-driver architecture with watchdog safety stops and per-node feedback for scalable wireless actuation experiments.

CLD Passive Impedance Shaping and RL Exploration

- Led experimental validation of constrained-layer damping as passive impedance shaping for soft robotic swimmers, achieving about 5x peak acceleration and 3x terminal velocity over baseline.
- Characterized complex bending impedance of CLD sandwich structures using torque sensing, dry bender tests, lock-in harmonic regression, constrained propulsion, and phase-resolved PIV.
- Designed a MuJoCo/Gymnasium RL framework where frequency-dependent material impedance reduces swimming control to frequency and amplitude while improving sim-to-real robustness.

Flow Sensing and ML-Based Perception

- Built PyTorch sparse-to-full flow reconstruction models that infer velocity and vorticity structure from local PIV patches around a NACA 0012 foil.
- Designed vortex-focused metrics, physics-informed losses, and a dual-pathway PIV-plus-pressure architecture with attention and ensemble uncertainty.
- Redesigned data collection around Strouhal number, foil spacing, and chord variation, using active learning to prioritize informative wake-topology cases.

Heaving Foil Near a Free Surface: Experimental Hydrodynamics

- Executed heaving hydrofoil experiments near a deformable free surface using force sensing, PIV, potential-flow simulation, and wave-theory analysis.
- Identified vortex-surface phase coupling as the mechanism behind lift reversal, linking unsteady-number effects to vertical force generation.

- Processed experimental datasets in Python and MATLAB, supporting manuscript revision with PIV evidence, hydrodynamic decomposition, and free-surface validation.

ADDITIONAL ENGINEERING PROJECTS

Applied PIV Collaboration with John Deere

- Supported applied PIV experiments for engineering flow analysis, contributing imaging setup, data processing, and interpretation for industry-facing measurements.
- Translated experimental flow diagnostics into actionable observations, strengthening practical test planning and communication with external collaborators.

ClearCoach: Computer Vision Video Analysis Tool

- Built and deployed a Flask/OpenCV web application that identifies five badminton stroke phases from uploaded clips using LLM-vision APIs.
- Designed a contact-sheet inference pipeline with structured JSON outputs, showing practical full-stack deployment, vision preprocessing, and prompt-based model integration.

SELECTED PUBLICATIONS

Feng, Q., Roberts, O. A., and Zhong, Q. "Passive Phase-Oriented Impedance Shaping for Rapid Acceleration in Soft Robotic Swimmers." arXiv:2603.03537, submitted to IROS 2026.

Feng, Q., Han, T., and Zhong, Q. "Lift reversal from vortex-surface phase coupling in a heaving foil near a free surface." arXiv:2512.12485, 2025.

Zhu, Y., Liu, L., Han, T., Feng, Q., Moored, K. W., Zhong, Q., and Quinn, D. B. "Wavenumber affects the lift of ray-inspired fins near a substrate." Journal of the Royal Society Interface, 22:20250276, 2025.

Feng, Q. and Zhong, Q. "Mantabot Swimming Performance." Manuscript in preparation.

HONORS AND PRESENTATIONS

Link Foundation Ocean Engineering and Instrumentation Fellowship | Jul 2024

Presentation, Annual Meeting of the APS Division of Fluid Dynamics | Nov 2024

Presentation, CoMFRE Annual Meeting, Iowa State University | Oct 2024

National Undergraduate Academic Science and Technology Competition | Oct 2021